MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 5684

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, inert gas

See figure 1

Mounting position: Any

Weight: 6-ounces (170.1 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	epy	Ib	ib	i surge 0.1 sec	tk	TA
Unit:	Vac	v	v	Adc	a	a	sec	°C
Maximum:	2.63	1,250	1,000	2.5	30	300	---	75
Minimum:	2.37	---	---	---	---	---	30	-55
TEST CONDITIONS:	2.5	1,250	1,000	---	---	---	---	---

GENERAL

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3201	Critical grid voltage for conduction	-	Ebb = 1,000 Vdc; Rg = 1,000 ohms; Rp = 750 to 5,000 ohms	0.65	II	Ecc	-4.2	-7.8	Vdc
(D) 3201	Critical anode voltage conduction	-	Ecc = -43 Vdc; Rg = 1,000 ohms; Rp = 750 to 5,000 ohms	0.65	II	Ebb	---	75	Vdc
(D) 3204	Voltage drop	2	Ib = 2.5 Adc; Epp = 130 Vac (max); Rg = 10,000 ohms; grid connected to anode	0.65	II	Etd	---	12	Vdc
1231	Peak emission by voltage drop	1	ib = 30 a; grid connected to anode thru 10,000 ohms	0.65	II	etd	---	35	v
(D) 3241	Heater current	-		0.65	II	If	7.0	11.0	Aac
	Quality conformance inspection, part 2								
(D) 3216	Grid current	4	Rg = 1 Meg and 10,000 ohms	---	---	Ic	---	5.0	uAdc
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
1036	Bump	-	Angle = 20°	---	---	---	---	---	---
3206	Operation	-	Ib = 2.5 Adc; Rg = 10,000 ohms	---	---	Ecc	-4.2	-7.8	Vdc
	Quality conformance inspection, part 3								
---	Life test (1)	3	Group D; operation; ib = 30 a; t = 500 hours	---	---	---	---	---	---
(D) ---	Life test (2) (no-load operation)	5	Group A; Ef = 2.5 V (no other voltages); t = 100 hours	---	---	---	---	---	---
---	Life-test (1) and (2) end points:								
1231	Peak emission by voltage drop	-		---	---	etd	---	50	v
3216	Grid current	-		---	---	Ic	---	10	uAdc
3206	Operation	-		---	---	Ecc	-4.2	-7.8	Vdc

NOTES:

1. This test is to be the first test performed at the conclusion of the holding period.
2. Wattmeter method: With an alternating current source and resistive load, measure the power loss in the tube with a wattmeter so that the current coil is in series with, and the voltage coil across, the tube. The tube voltage drop (Etd) is the total power loss minus the coil loss under load, divided by the average, or dc, current (Ib).

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NOTES: -Continued

(D) 3. Life-test operation may be at an optional Epp = 220 Vac (minimum) provided the required values of anode current (peak and average) are obtained.

4. Operate tube at least 5 minutes before the test. Operation may be at optional Epp, but Ib must be 2.5 Adc minimum. To test, switch to the test voltage, and Ib = 2.5 Adc in less than 2 seconds.

5. After no-load operation, tubes shall be within specification limits for operation and grid-current tests. For operation test, after no-load operation life test, the operating time shall be a minimum of 30 and a maximum of 120 seconds.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

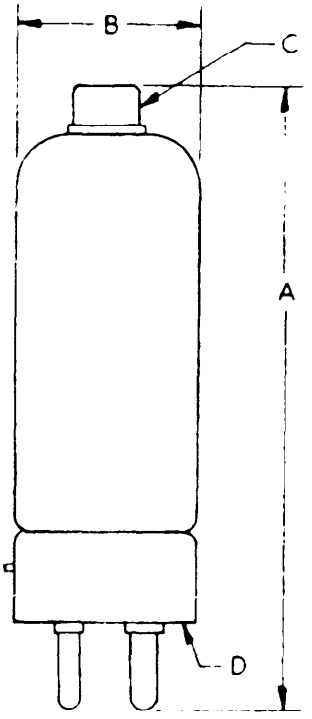
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2926)

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PIN CONNECTIONS	
Pin No.	Element
1	f
2	fct
3	g
4	f
cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	---	6.250 (158.75)
B	1.438 (36.53)	1.688 (38.86)
Ⓓ Quality conformance inspection, part 3 (see note)		
C	Cap: C1-5 (EIA)	
D	Base: A4-10 (EIA)	

NOTE:
This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 5684.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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